Mono Audio Equalizer

Professional Audio Processing Module

1. Product Overview



The mono audio equalizer allows precise control over low, mid, and high frequency bands. It supports multiple audio inputs over microphone or instrument and delivers clear, low-distortion output. Designed for professional audio applications.

2. Key Features

- Mono audio processing
- High SNR (>80 dB)
- Wide frequency range: 20 Hz – 20 kHz
- Buffered inputs and low-impedance output
- Compact PCB design with standard mounting points

3. I/O Specification

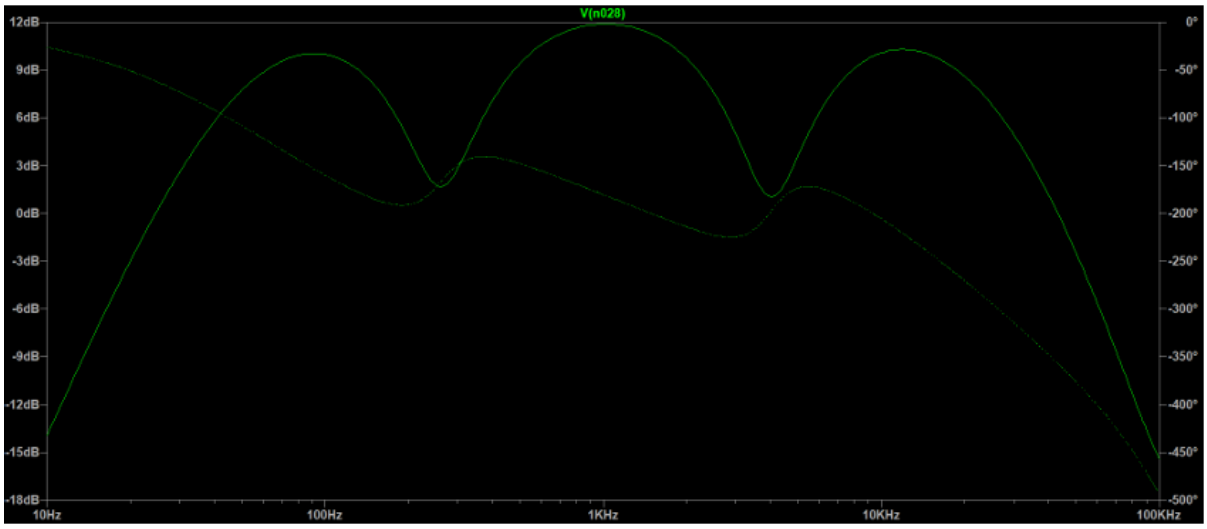


Figure 1: Equalizer Frequency response.

I/O Name	Mode	Unit	Specification		
			Min	Typ	Max
Input 1	Mic	mV	2	6	10
	Instrument	mV	100	150	200
Input 2	Mic	mV	2	6	10
	Instrument	mV	100	150	200
Input 3	Mic	mV	2	6	10
	Instrument	mV	100	150	200
Output	-	Vpp	0	1	2
Power Supply	Voltage	V	216	230	244
	Current	mA	50	65	100
Output Power	-	W	0.1	0.5	1

Table 1: mono audio I/O specifications

4. Device Specification

Parameter	Unit	Rating		
		Min	Typ	Max
Supply Voltage	V	216	230	244
Supply Current	A	—	1	1.5
Input Impedance	kΩ	3	3.5	5
Output Impedance	Ω	—	8	—
SNR	dB	80	85	—
Frequency Range	Hz	50	10000	20000

Table 2: Device electrical specifications.

5. Control Specification

Control Parameter	Unit	Range		
		Min	Typ	Max
Low Band (50–250 Hz) Gain	dB	-12	0	+12
Mid Band (250 Hz – 4 kHz) Gain	dB	-12	0	+12
High Band 4–20 kHz) Gain	dB	-12	0	+12

Table 3: Control ranges for each frequency band.

6. Enclosure & Interface



Figure 2: Device dimentions

- **matterial :** Polylactic Acid (PLA)
- **wall thickness :** 3 mm (high strength)

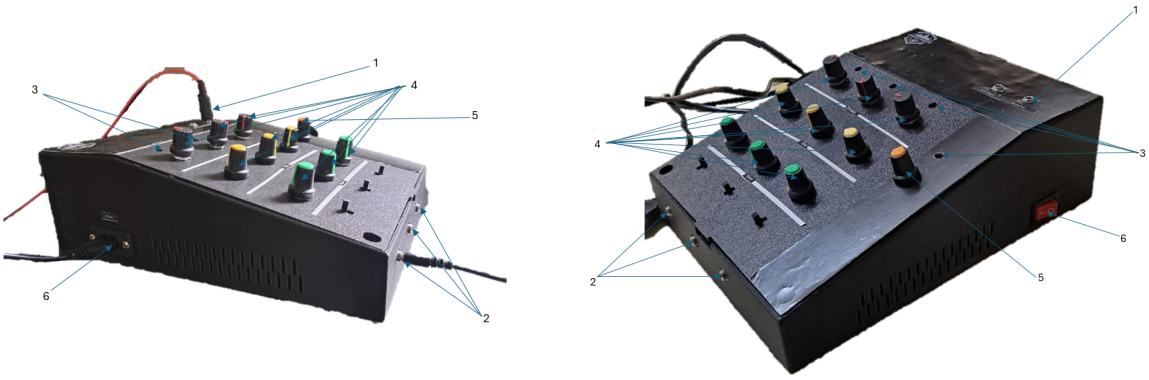


Figure 3: enclosure

1. AUX Out (3.5mm jack)
2. AUX In (3.5mm jack)
3. LED Displays
4. Individual Gain Controllers
5. Volume Controller
6. Power

7. Block Diagram

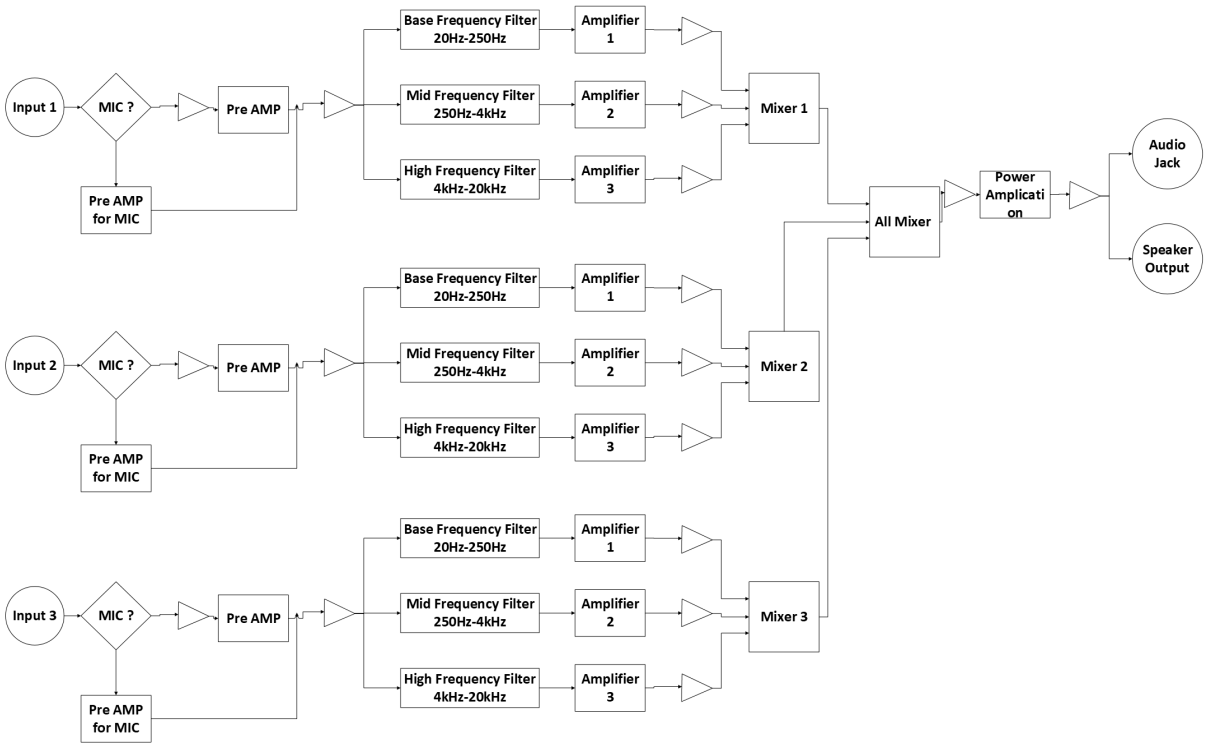


Figure 4: System Architecture

8. PCB Schematics

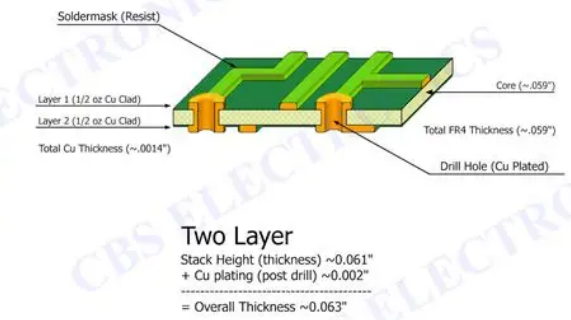


Figure 5: PCB Layout

8.1 Power Supply Section

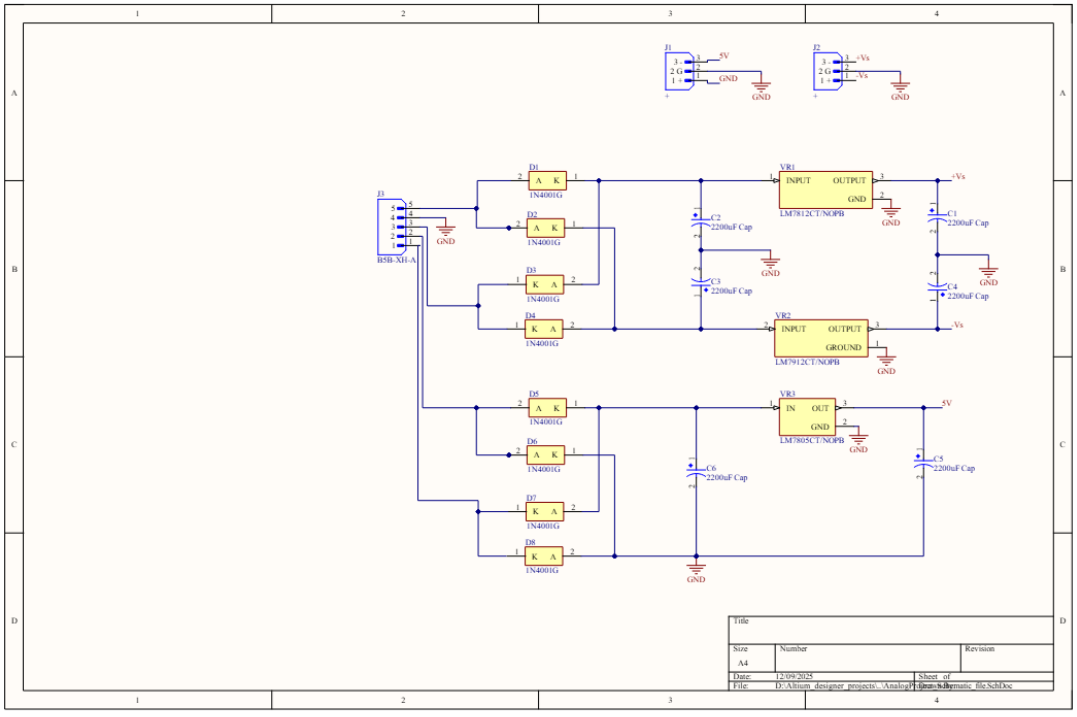


Figure 6: Power supply schematic

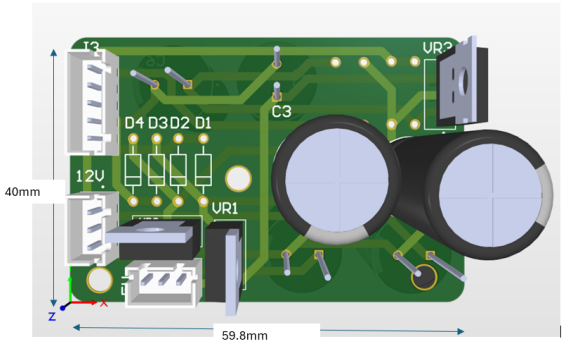


Figure 7: Power supply PCB

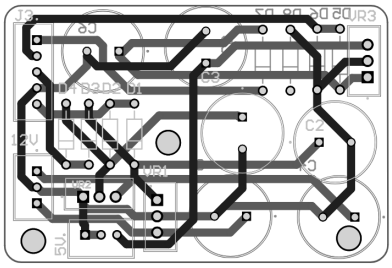


Figure 8: Power supply traces

8.2 Equalizer Section

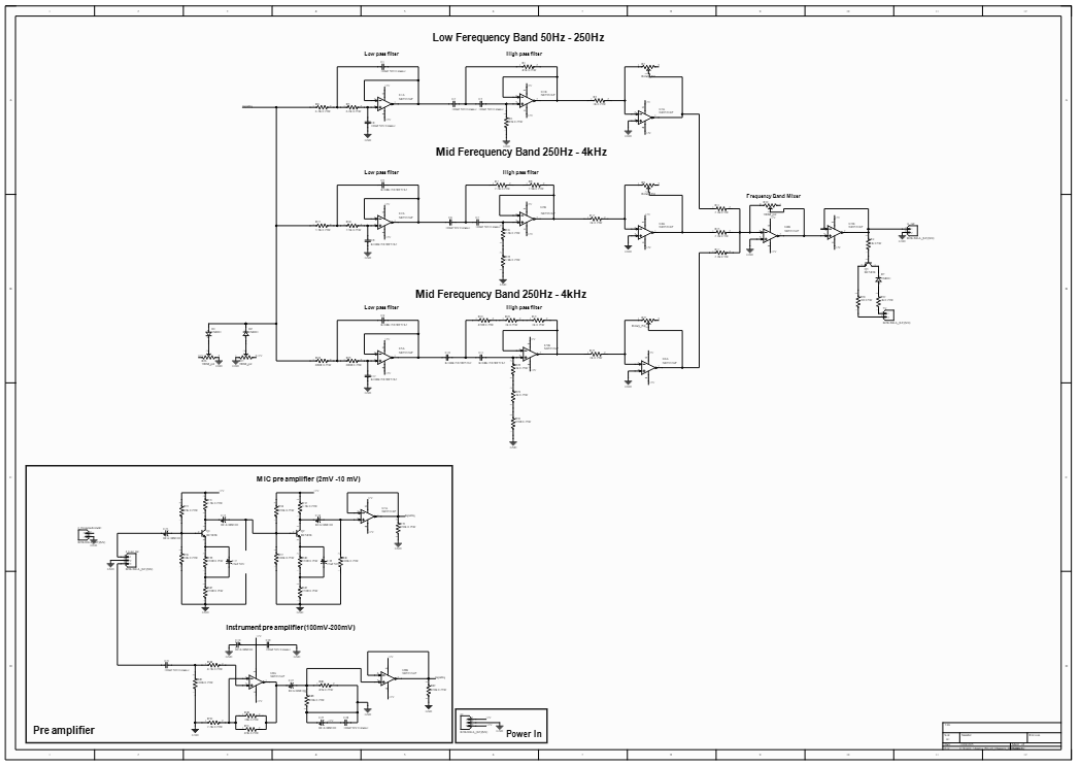


Figure 9: Equalizer schematic

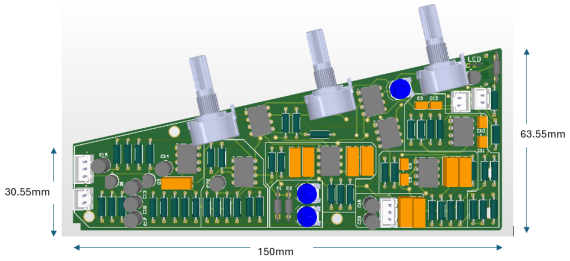


Figure 10: Equalizer PCB

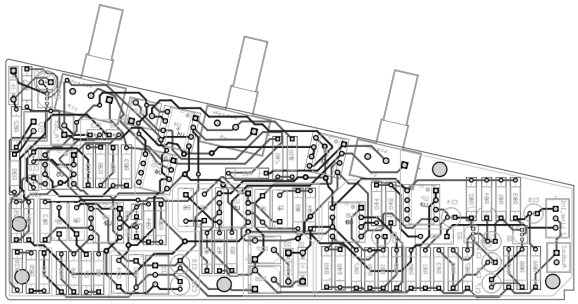


Figure 11: Equalizer traces

8.3 Mixer Section & Power Amplifier Section

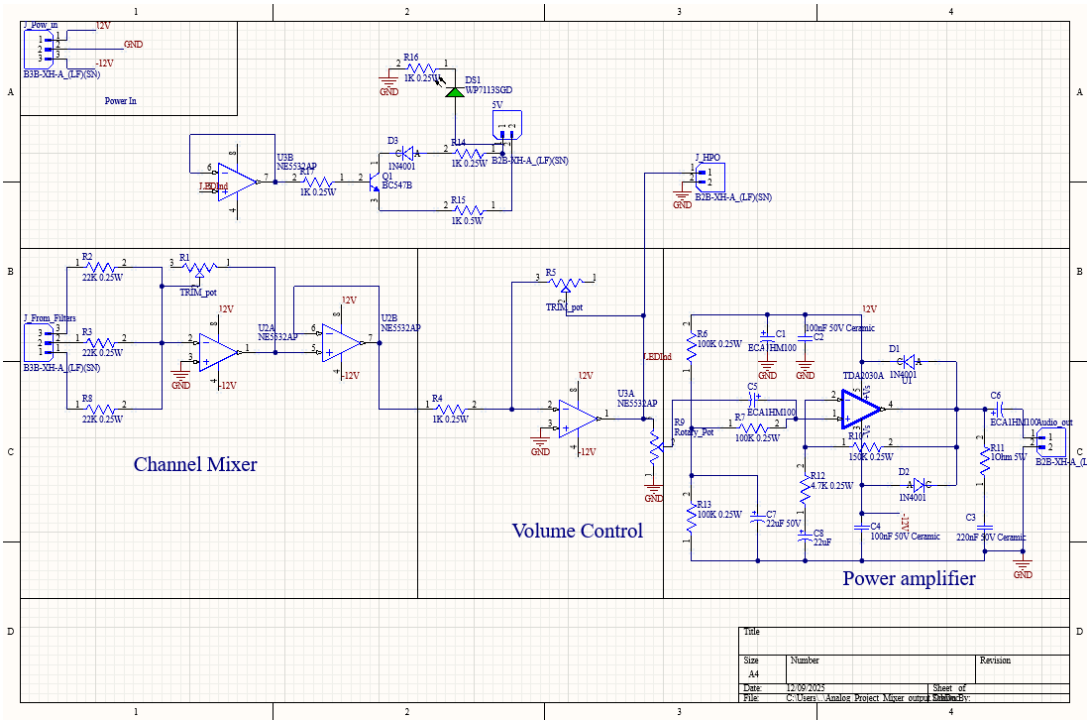


Figure 12: Mixer schematic

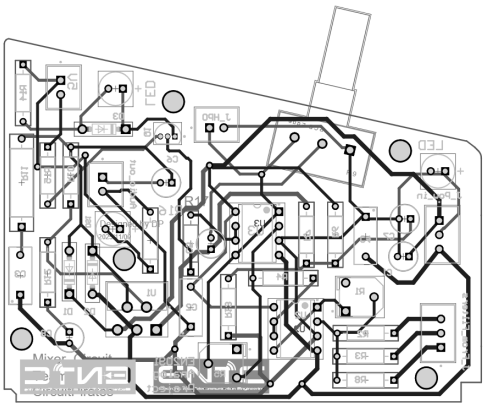


Figure 13: Mixer traces

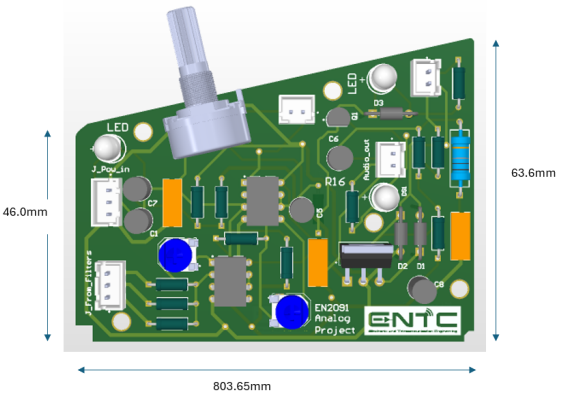


Figure 14: Mixer output PCB